

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1722

Speed 138 kV
March 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1722. The Interconnection Customer proposes a 126.0 MW battery energy storage facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on April 17, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 622 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,779,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
56Th St / Oates Reit 138kV	Speed / Silver Bay Power 138kV L	412.2	247	100.0
56Th St / Santa Monica 138kV	Silver Bay Power / Speed 138kV	352.9	298	100.0
Brockport / Narod Substation 138kV	Silver Bay Power Substation / Speed 138kV Line	348.4	305	100.0
Santa Monica / Hambrick 138kV L	Silver Bay Power / Speed 138kV	334.2	321	100.0
Barnham / Narod 138kV	Silver Bay Power / Speed 138kV	319.1	344	100.0
Gaga / Barnham 138kV line	Silver Bay Power / Speed 138kV	304.9	367	100.0
Speed / Silver Bay Power Substation 138kV	Speed / Gaga 138kV	292.8	388	100.0
Speed / Gaga 138kV L	Silver Bay Power Substation / Speed 138kV	292.8	388	100.0
Silver Bay Power / Hotel Rd Substation 138kV line	Speed / Gaga 138kV L	174.8	266	31.8
Silver Bay Power Substation / Damon 138kV	Silver Bay Power / Hotel Road 138kV line	168.5	230	33.9
Stelco / Murphys 138kV	Pine Bluff Watson Chapel / Carraizo 1201 345kV Line	126.7	294	14.9
Murphys / Culver 138kV line	Pine Bluff Watson Chapel / Carraizo 1201 345kV	115.5	210	9.6
Stelco / Dewitt 138kV	Pine Bluff Watson Chapel / Carraizo 1201 345kV line	112.9	204	7.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Greenville Ave / Valliant 138kV	Hotel Rd / Silver Bay Power 138kV	106.3	206	19.8
Cpu Silicon Substation / Lanham 138kV Line	Cooper / Lee County Solid Waste Energy 345kV line	102.7	175	6.2
Eldorado / Action 138kV	Carraizo 1201 / Pine Bluff Watson Chapel 345kV	102.4	638	32.9
Action / Alibates 138kV	Pine Bluff Watson Chapel / Carraizo 1201 345kV Line	101.7	682	32.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
56Th St / Oates Reit 138kV	Rebuild line with higher-capacity conductor	\$2,877,000	26% (\$751,000)
56Th St / Santa Monica 138kV	Rebuild line with higher-capacity conductor	\$2,341,000	26% (\$611,000)
Brockport / Narod Substation 138kV	Rebuild line with higher-capacity conductor	\$4,319,000	26% (\$1,128,000)
Santa Monica / Hambrick 138kV line	Rebuild line with higher-capacity conductor	\$4,707,000	26% (\$1,229,000)
Barnham Substation / Narod 138kV line	Rebuild line with higher-capacity conductor	\$8,544,000	26% (\$2,231,000)
Gaga / Barnham 138kV	Rebuild line with higher-capacity conductor	\$1,994,000	26% (\$521,000)
Speed / Silver Bay Power Substation 138kV	Rebuild line with higher-capacity conductor	\$18,021,000	26% (\$4,706,000)
Gaga / Speed 138kV	Rebuild line with higher-capacity conductor	\$2,317,000	26% (\$605,000)
Silver Bay Power / Hotel Rd 138kV Line	Rebuild line with higher-capacity conductor	\$14,200,000	15% (\$2,131,000)
Silver Bay Power Substation / Damon 138kV	Rebuild line with higher-capacity conductor	\$3,866,000	24% (\$940,000)
Murphys / Stelco 138kV	Rebuild line with higher-capacity conductor	\$6,636,000	7% (\$474,000)
Culver / Murphys 138kV	Rebuild line with higher-capacity conductor	\$11,001,000	7% (\$813,000)
Stelco / Dewitt 138kV	Rebuild line with higher-capacity conductor	\$14,376,000	8% (\$1,084,000)
Greenville Ave / Valliant 138kV line	Reconductor line	\$3,564,000	18% (\$654,000)
Cpu Silicon Substation / Lanham 138kV Line	Reconductor line	\$1,054,000	5% (\$51,000)
Eldorado / Action 138kV	Reconductor line	\$1,137,000	8% (\$91,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Action / Alibates 138kV	Reconductor line	\$9,523,000	8% (\$759,000)
	Total allocated to project		\$18,779,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.